

Omar Khan

Ph.D. Candidate in Human-Computer Interaction

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Research Interests

Mixed-methods UX and HCI researcher at the intersection of accessibility, visualization, and AI-assisted content creation. Building AI systems democratizing qualitative visualization authoring, enabling users with diverse abilities to create interpretive diagrams and conceptual models independently. Research with blind and low-vision scientists informs transferable methods for inclusive visual communication across STEM, education, and collaborative knowledge sharing.

Education

University of Illinois Urbana-Champaign

Ph.D., Computer Science 2022 - 2027
Advisor: Dr. JooYoung Seo
Certificate in Information Accessibility Design and Policy

University of Illinois Urbana-Champaign

B.S., Computer Science + Crop Sciences 2017 - 2021

Skills

Programming Languages	Python ^{***} , Java ^{**} , JavaScript ^{**} , R ^{**} , Swift ^{**} , C++ [*]
Technologies	HTML ^{***} , CSS ^{***} , Git ^{**} , LaTeX ^{**} , MAXQDA ^{**} , MS Office ^{**} , Qualtrics ^{**} , SQL ^{**} , MongoDB [*] , Neo4J [*] , Docker [*] , Kubernetes [*] , Figma [*]
Frameworks	React ^{**} , Next.js ^{**} , PyTorch ^{**} , SciPy ^{**} , TensorFlow ^{**} , scikit-learn ^{**} , SciPy ^{**} , Matplotlib ^{**} , NumPy ^{**} , Pandas ^{**} , NLTK ^{**} , SQLAlchemy [*] , Flask [*] , d3.js [*] , Hadoop [*] , Spark [*]
Research Methods	Interview Design ^{***} , Semi-Structured Interviews ^{***} , Thematic Analysis ^{***} , A/B Testing ^{**} , Co-Design ^{**} , Experimental Design ^{**} , Prototyping ^{**} , Statistical Analysis ^{**} , Survey Design ^{**} , Usability Testing ^{**} , Affinity Diagrams [*] , Heuristic Evaluation [*]

^{*} = Familiar; ^{**} = Experienced; ^{***} = Proficient

Research Experience

(x)Ability Design Lab Champaign, IL
Graduate Research Assistant | Advisor: Dr. JooYoung Seo 05/2024--Present

- **QUARTZ (Qualitative Understanding via Accessible Representation and Visualization)**
 - Leading user research to understand barriers in qualitative data visualization, conducting **15+** co-design sessions
 - Designing user study protocols, including survey instruments, semi-structured interview protocols, and task-based usability testing
 - Managing global participant recruitment pipeline via BLV advocacy networks (e.g., World Blind Union)
- **Re-Imagining Knowledge Production Workflows by, with, and for BLV Researchers**
 - Led a mixed-methods global study with BLV researchers (survey = 57; interviews = 15) using activity theory to examine research tool barriers, revealing **17% task delegation rate** and critical gaps in data visualization tools (**2.58/5 difficulty**).
 - Synthesized findings through reflexive thematic analysis to develop systemic design recommendations addressing workflow contradictions across literature review, analysis, and dissemination stages.
- **LLM Fine-Tuning for Accessibility Applications Using High-Performance Computing**
 - Fine-tuning large language models on distributed GPU clusters to develop accessibility-focused AI solutions, optimizing performance for assistive technologies.
- **Accessible Extended Reality (XR) Co-Design**
 - Co-designed *PunchPulse*, an accessible VR boxing game with a **5-person mixed-abilities team**, leveraging the Rapid Iterative Testing and Evaluation (RITE) framework over 16 weeks
- **Multimodal LLM-Based Data Visualization**
 - Collaborated with **8 developers** and **3 researchers** to build, refine, and advance MAIDR, a multimodal AI-driven system for accessible data visualization.

CROPS Lab

Graduate Research Assistant | Advisor: Dr. Camille Cobb

Urbana, IL
08/2022--05/2024

- **YouTube Influencer Ads and Privacy Promises**
 - Led large-scale quantitative analysis of **300M YouTube influencer privacy technology advertisements**, identifying top-advertised products and consumer misinformation risks.
- **Smart Home Data Collection Consent Dynamics**
 - Co-led mixed-methods analysis of **360-participant survey** examining consent dynamics in smart home environments, revealing **84% of users experienced consent violations**.
 - Developed stakeholder-specific recommendations for consentful smart home ecosystems.

National Center for Supercomputing Applications

Research Assistant | Advisors: Dr. Megan Dailey & Dr. Matthew Hudson

Urbana, IL
08/2021--12/2021

- **Food Bank Marketplace Mobile App**
 - Developed the initial software design specification for a mobile app aimed to create a marketplace to provide food banks with the means to procure fresh produce from farmers at a reduced cost, aiming to reduce food waste and insecurity.

ORCHID Lab

Undergraduate Research Assistant | Advisor: Dr. Brian P. Bailey

Urbana, IL
01/2021--05/2021

- **Proximal Process Goal Accomplishment**
 - Researched methodologies and visualizations to help individuals accomplish proximal process goals (goals that are achievable in a short amount of time and involve specific processes).

Siebel School of Computing and Data Science

Undergraduate Research Assistant | Advisor: Dr. Lawrence C. Angrave

Urbana, IL
01/2021--05/2021

- **Webcam-Based Eye Tracking in ClassTranscribe**
 - Investigated the challenges of webcam-based eye tracking and an implementation of it in ClassTranscribe, an online lecture streaming platform at the University of Illinois, using WebGazer, a React.js-based framework developed by Brown University.

Siebel School of Computing and Data Science

Undergraduate Research Assistant | Advisor: Dr. Geoffrey L. Herman

Urbana, IL
05/2019--08/2019

- **Computerized Exam for Computer Architecture**
 - Designed, implemented, and deployed a computerized, multi-part exam with another undergraduate student on pipelining and cache analysis for CS 233: Computer Architecture, taken annually by **400+ students**, introducing hardware concepts
 - Co-authored a publication on the implications of our implementation on engineering education beyond computing

Professional Experience

Infosys

Research Scientist Intern (Center for Autonomous Machines)

Remote
06/2026-Present

- Leading an **exploratory sequential mixed-methods study (online focus groups + survey)** to surface inclusive design principles for autonomous last-mile delivery robots (LMDRs)

Fable

Accessibility Tester (Contract)

Remote
09/2025-Present

- Conducting accessibility testing (user interviews, prototype reviews, and compatibility assessments) for a wide range of clients across multiple industry domains through the U.S, Canada, and the U.K.
- Partnering with cross-functional teams to translate findings into actionable, prioritized recommendations for inclusive design

American Foundation for the Blind (AFB)

Researcher (Volunteer)

Remote
09/2025-Present

- Conducting comparative thematic analysis across disabled and non-disabled survey respondents to characterize differential experiences with AI systems; identified key accessibility gaps and designed evidence-based interventions

Viasat

Software Engineer Intern

Carlsbad, CA

05/2021-08/2021

- Designed and implemented a Slack app with 2 other interns using Python (frontend), SQLAlchemy, and Kubernetes (backend) that allows users to manually archive older Jira projects and automatically archive them by a given date

L'Oreal U.S.A.

Software Engineer Extern

Remote

06/2020-08/2020

- Collaborated with a team of 4 externs in a product design competition, implementing a data-driven solution to reduce the amount of plastic used in L'Oreal product packaging

Publications and Presentations

Conference and Journal Publications

- (C1) **Omar Khan** and JooYoung Seo. 2026. QUARTZ: Qualitative Understanding via Accessible Representation and Visualization. *To appear in the 28th International ACM SIGACCESS Conference on Computers and Accessibility, Porto, Portugal*. ACM. [\[TBA\]](#)
- (C2) **Omar Khan** and JooYoung Seo. 2026. "I Don't Want My Mental Health App To Give Me Mental Health Barriers": Unpacking The Need For Digital Mental Health Tracking Services With And For The Blind Community. *To appear in the 28th International ACM SIGACCESS Conference on Computers and Accessibility, Porto, Portugal*. ACM. [\[TBA\]](#)
- (C3) **Omar Khan** and JooYoung Seo. 2026. "I Don't Trust Any Professional Research Tool": A Re-Imagination of Knowledge Production Workflows by, with, and for Blind and Low-Vision Researchers. *In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. April 13--17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 24 pages. [\[DOI\]](#) [\[arxiv\]](#)
- (C4) Sanchita S. Kamath, **Omar Khan**, Anurag Choudhary, Jan Meyerhoff-Liang, Soyoung Choi, JooYoung Seo. 2025. PunchPulse: A Physically Demanding Virtual Reality Boxing Game Designed with, for and by Blind and Low-Vision Players. *In The 27th International ACM SIGACCESS Conference Computers and Accessibility (ASSETS '25)*, October 26--29, 2025, Denver, CO, USA. ACM, New York, NY, USA, 21 pages. [\[DOI\]](#)
- (C5) Sanchita S. Kamath, **Omar Khan**, Aziz N. Zeidieh, and JooYoung Seo. 2025. Sensing the Shape of Data: Non-Visual Exploration of Statistical Concepts in Histograms with Blind and Low-Vision Learners. *arXiv preprint arXiv:2509.14452*. [\[arxiv\]](#)
- (C6) Alex Atcheson, **Omar Khan**, Brian Siemann, Anika Jain, and Karrie Karahalios. 2025. "I'd Never Actually Realized How Big An Impact It Had Until Now": Perspectives of University Students with Disabilities on Generative Artificial Intelligence. *In CHI Conference on Human Factors in Computing Systems (CHI '25)*, April 26--May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 22 pages. [\[DOI\]](#)
- (C7) Yi-Shyuan Chiang, **Omar Khan**, Adam Bates, and Camille Cobb. (2024). More than just informed: The importance of consent facets in smart homes. *In Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*, May 11--16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 21 pages. [\[DOI\]](#)
- (C8) Suleman Mahmood, Mingjie Zhao, **Omar Khan**, Geoffrey L. Herman. (2020). Caches as an Example of Machine-gradable Exam Questions for Complex Engineering Systems. *In Proceedings of the 49th ASEE/IEEE Frontiers in Education Conference, Uppsala, Sweden, Oct. 21--24*. [\[DOI\]](#)

Public Reports & White Papers

- (R1) Arielle M. Silverman, Angie L. Whistler, Alyssa Shock, Carmel H. Heydarian, Sarahelizabeth J. Baguhn, William E. Hanuschock, III, Mana Hashimoto, **Omar Khan**, & Mei-Lian Vader. (2026). The AI Quagmire: Benefits, Risks, and User Aspirations Through a Disability Lens. *American Foundation for the Blind*. [\[Link\]](#)

Works in Progress, Posters, and Presentations

- (W1) **Omar Khan** and JooYoung Seo. 2025. Sighted People Have Their Pick Of The Litter": Unpacking The Need For Digital Mental Health (DMH) Tracking Services With And For The Blind Community. *In Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26--May 1, 2025, Yokohama, Japan. ACM, New York, NY, USA, 13 pages. [\[DOI\]](#)
- (W2) Sanchita S. Kamath, Aziz Zeidieh, **Omar Khan**, Dhruv Sethi, and JooYoung Seo. 2024. Playing Without Barriers: Crafting Playful and Accessible VR Table-Tennis with and for Blind and Low-Vision Individuals. *In Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24)*. Association for Computing Machinery, New York, NY, USA, Article 88, 1--5. [\[DOI\]](#)
- (W3) Harsh Deep, Joanna Huang, Rashi Dembi, **Omar Khan**, and Lawrence Angrave. Webcam Eye-Tracking Based Accessibility for ClassTranscribe. *Virtual poster presentation: Illinois Undergraduate Research Symposium 2021*.

Service and Leadership

RESEARCH SERVICE

2026

Technical Paper Reviewer

DIS 2026

COMMUNITY SERVICE

2026-Present	Vice President, Board of Directors PACE - Center for Independent Living	Urbana, IL
2024-2025	Broadening Participation in Computing (BPC): K-12 Student Outreach Committee	UIUC
2023-Present	Director, Board of Directors PACE - Center for Independent Living	Urbana, IL
2021-2022	Campus Co-Director, Fundraising Team The Percentage Project Inc.	Remote
2020-2022	Facilitator, Curriculum Development Co-Lead Girls Who Code	UIUC
2021	Mentoring Co-Chair Illinois Women in Computer Science	UIUC
2019-2020	Content Co-Chair Reflections Projections	UIUC
2019-2020	Co-Director Illinois Computer Science Sail	UIUC

Teaching Experience

S24	Graduate Teaching Assistant: CS 467: Social Visualization	UIUC
F23, F24	Graduate Teaching Assistant: CS 210/211: Ethical & Professional Issues/Conduct in CS	UIUC
F19, S20, F20, S21	Undergraduate Course Assistant: CS 233: Computer Architecture	UIUC
F18, S19, F19	Undergraduate Course Assistant: CS 196-25: Freshman Honors	UIUC
F18, S19	Undergraduate Course Assistant: CS 125: Introduction to Computer Science	UIUC

Mentoring

2025-Present	Soundarya Kumar Pradhan	UIUC MCS
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Awards and Honors

08/2025	SIGACCESS Travel Scholarship	ACM
12/2024	2025 CRA-WP Grad Cohort for Women & Grad Cohort for IDEALS	CRA-WP
04/2021	Most Long-Term Social Impact: Stanford Open Datathon 2021	Stanford
2020-2021	The Climate Corporation Scholarship	UIUC